

Vimal Manoharan

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EDUCATION

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science Engineering (CGPA: 8.91/10.0)

Chennai, India

Sep. 2022 – May 2026

EXPERIENCE

Aakar Labs

React Native Frontend Developer Intern

Chennai, India

Sep. 2025 – Mar. 2026

- Collaborated within an Agile, product-focused team environment to design and pair-program iterative mobile interfaces for Aaku, an AI-powered travel companion utilizing React Native and Expo.
- Partnered closely with backend AI engineers to assist in the development and validation of an AI agent trip planning feature, ensuring dynamic frontend rendering of multi-agent execution workflows.
- Applied structured Object-Oriented patterns to engineer reusable UI components and state-driven flows across planning, trip progression, and contextual journey interaction modules.
- Integrated REST APIs and optimized frontend rendering performance, lowering end-user latency to support seamless, real-time client itinerary updates.
- Engaged in regular cross-functional feedback sessions with backend engineers and product designers to iteratively improve multi-step planning architectures and UX consistency.

KSK Electronics Pvt. Ltd.

Software Developer Intern

Chennai, India

Aug. 2025 – Oct. 2025

- Developed and refactored a full-stack ERP platform using React, Node.js, and MongoDB, migrating business-critical financial records from Zoho SaaS through clean, maintainable code architectures.
- Architected and deployed a corporate RAG (Retrieval-Augmented Generation) SOP framework using LangChain and Chroma DB, allowing employees to instantly retrieve exact policy explanations and FAQ answers.
- Implemented Object-Oriented design and role-based access control (RBAC), optimizing backend database queries to reduce stock retrieval latency.
- Established Continuous Integration (CI) concepts by containerizing services with Docker and deploying onto AWS EC2 with Nginx, significantly advancing delivery reliability and platform uptime.
- Followed test-focused logic to automate GST validation workflows, eliminating manual accounting and data-entry discrepancies by 90% during monthly audit operational cycles.

Siddha Shivalayas Clinic

Freelance Full-Stack Developer

Chennai, India

Jan. 2025 – Apr. 2025

- Listened thoughtfully to client business goals to design and iteratively build a MERN-stack Clinic Management System managing patient records, billing, and inventory workflows for 300+ active profiles.
- Adhered to SOLID principles to implement real-time inventory deduction logic, ensuring complete transactional database consistency during concurrent checkout operations.
- Built analytics dashboards using Material UI to map patient histories and financial trends, removing administrative bottlenecks and translating complex data into clear stakeholder value.

PROJECTS

Agentic Market Intelligence Engine | *Java, Spring Boot, LangChain, Gemini API, Chroma DB, Selenium*

- Built a multi-threaded web scraping backend utilizing Java ExecutorService to concurrently pull marketplace product datasets under robust OO-design principles.
- Leveraged LangChain workflows to deploy automated semantic analysis agents, transforming unstructured multi-channel customer reviews into structured database insights.
- Integrated Graph RAG and Chroma vector databases to optimize cross-domain entity matching, accelerating lookups for competitive product datasets.
- Utilized Test-Driven Development (TDD) philosophies to build and evaluate Jaro-Winkler fuzzy matching algorithms, mapping item variations across competitive domains with 95% matching accuracy.

Explainable AI Framework for Early Risk Stratification | *Python, FastAPI, Scikit-learn, SHAP, NumPy*

- Designed and implemented an iterative predictive healthcare system utilizing ensemble machine learning models to run early clinical risk assessment and disease diagnosis.
- Engineered a probabilistic multimodal fusion pipeline to aggregate patient history and clinical metrics, translating unstructured medical data into predictive insights using clean analysis and coding principles.
- Integrated Explainable AI (XAI) frameworks utilizing SHAP and counterfactual analysis to extract feature importance, ensuring clinical transparent reasoning for diverse stakeholder perspectives.
- Built and deployed a low-latency FastAPI backend service structured to deliver real-time diagnostic predictions and visualization payloads securely.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL, HTML, CSS

Frameworks: React.js, React Native, Expo, Node.js, Express.js, Spring Boot, FastAPI, LangChain, LangGraph, RAG, Tailwind CSS, Material UI

Developer Tools: Git, Docker, AWS EC2, Nginx, Chroma DB (Vector DB), GitHub, Android Studio, VS Code

Methodologies & Libraries: AI Agents, RAG Pipelines, TDD, Agile, OOP, SOLID, SHAP, Pandas, NumPy, Scikit-learn, Selenium